

more general model-theoretic point of view, and have therefore been based on the strongest of all graph relations, the induced subgraph relation. As a consequence, most of these results are negative; see the notes.

From a graph-theoretic point of view, it seems more promising to look instead for universal graphs for the weaker subgraph relation, or even the topological minor or minor relation. For example, while there is no universal planar graph for subgraphs or induced subgraphs, there is one for minors:

Theorem 8.3.4. (Diestel & Kühn 1999)

There exists a universal planar graph for the minor relation.

This remained the only result about universal graphs for the minor relation for over 20 years, until Georgakopoulos found $\preceq$ -universal graphs in $\text{Forb}_{\preceq}(X) = \{G \mid X \not\preceq G\}$ for $X = K^5$ and $X = K_{3,3}$, and proved there is none for $X = K^{\aleph_0}$.

8.4 Connectivity and matching

In this section we look at infinite versions of Menger's theorem and of the matching theorems from Chapter 2. This area of infinite graph theory is one of its best developed fields, with several deep results. One of these, however, stands out among the rest: a version of Menger's theorem that had been conjectured by Erdős decades ago, and was proved only fairly recently by Aharoni and Berger. The techniques developed for its proof inspired, over the years, much of the theory in this area.

Before we turn to this result, however, let us take a brief look at edge-connectivity. Recall from Section 8.1 that in an infinitely edge-connected countable graph we can easily find infinitely many edge-disjoint spanning trees. Can we still find such trees when the graph is uncountable? We can, but this is not quite as easy to prove (Exercise 62).

The following deep theorem of Laviolette reduces the above problem to its countable case – as it does for many other problems involving edge-connectivity. Let $\mathcal{H}$ be a set of countable graphs forming an edge-decomposition of an arbitrary graph G . Call this decomposition *bond-faithful* if every countable bond of G is contained in some $H \in \mathcal{H}$ and every finite bond of any $H \in \mathcal{H}$ is a bond also of G . Note that the finite bonds of G will be bonds of the $H \in \mathcal{H}$ that contain them. (Why?)

Theorem 8.4.1. (Laviolette 2005)

Every graph has a bond-faithful decomposition into countable graphs.

We shall not be able to prove Laviolette's theorem here. But let us illustrate its power in reducing problems to their countable case by deducing an early classic from the theory of infinite graphs.

even/odd
cut

Let us call a cut F in a graph G *even* or *odd* if $|F|$ is finite and even or odd, respectively. Since cycles meet cuts in an even number of edges, graphs with an odd cut cannot have an edge-decomposition into cycles. Surprisingly, all other graphs do:

Theorem 8.4.2. (Nash-Williams 1960)

Every graph with no odd cut has an edge-decomposition into cycles.

Proof. By Theorem 8.4.1 we may assume that our graph G is countable; let us enumerate its edges. We shall find the desired cycles in ω steps. At each step we shall assume inductively that every cut is infinite or even, find a cycle, and delete its edges.

To find that cycle, consider the first remaining edge $e = xy$ in our enumeration. By our inductive assumption e is not a bridge, so there is an x - y path not using e . Together with e this path forms a cycle, whose edges we delete to complete this step. This deletion keeps every finite cut even and every infinite cut infinite.

After at most ω steps no edges remain, so the cycles we found form an edge-decomposition of G . $\square$

Let us now turn to the Aharoni-Berger theorem: the infinite version of Menger's theorem originally conjectured by Erdős. We shall prove this theorem for countable graphs, which will take up most of this section. Although the countable case is much easier, the techniques it requires already give a good impression of the general proof. We then wind up with an overview of infinite matching theorems and a conjecture conceived in the same spirit.

Recall that Menger's theorem, in its simplest form, says that if A and B are sets of vertices in a finite graph G , not necessarily disjoint, and if $k = k(G, A, B)$ is the minimum number of vertices separating A from B in G , then G contains k disjoint A - B paths. (Clearly, it cannot contain more.) The same holds, and is easily deduced from the finite case, when G is infinite but k is still finite:

Proposition 8.4.3. *Let G be any graph, $k \in \mathbb{N}$, and let A, B be two sets of vertices in G that can be separated by k but no fewer than k vertices. Then G contains k disjoint A - B paths.*

(3.3.1) *Proof.* By assumption, every set of disjoint A - B paths has cardinality at most k . Choose one, $\mathcal{P}$ say, of maximum cardinality. Suppose $|\mathcal{P}| < k$. Then no set X consisting of one vertex from each path in $\mathcal{P}$ separates A from B . For each X , let P_X be an A - B path avoiding X . Let H be the union of $\bigcup \mathcal{P}$ with all these paths P_X . This is a finite graph in which no set of $|\mathcal{P}|$ vertices separates A from B . So $H \subseteq G$ contains more than $|\mathcal{P}|$ paths from A to B by Menger's theorem (3.3.1), which contradicts the choice of $\mathcal{P}$. $\square$

When k is infinite, however, the result suddenly becomes trivial. Indeed, let $\mathcal{P}$ be any maximal set of disjoint A – B paths in G . Then the union of all these paths separates A from B , so $\mathcal{P}$ must be infinite. But then the cardinality of this union is no bigger than $|\mathcal{P}|$. Thus, $\mathcal{P}$ contains $|\mathcal{P}| = |\bigcup \mathcal{P}| \geq k$ disjoint A – B paths, as desired.

Of course, this is no more than a trick played on us by infinite cardinal arithmetic: although, numerically, the A – B separator consisting of all the inner vertices of paths in $\mathcal{P}$ is no bigger than $|\mathcal{P}|$, it uses far more vertices to separate A from B than should be necessary. Or put another way: when our path systems and separators are infinite, their cardinalities alone are no longer a sufficiently fine tool to distinguish carefully chosen ‘small’ separators from unnecessarily large and wasteful ones.

To overcome this problem, Erdős suggested an alternative form of Menger’s theorem, which for finite graphs is clearly equivalent to the standard version. Recall that an A – B separator X is said to lie *on* a set $\mathcal{P}$ of disjoint A – B paths if X consists of a choice of exactly one vertex from each path in $\mathcal{P}$. The following so-called *Erdős-Menger conjecture*, now a theorem, influenced much of the development of infinite connectivity and matching theory:

*Erdős-
Menger
conjecture*

Theorem 8.4.4. (Aharoni & Berger 2009)

Let G be any graph, and let $A, B \subseteq V(G)$. Then G contains a set $\mathcal{P}$ of disjoint A – B paths and an A – B separator on $\mathcal{P}$.

The next few pages give a proof of Theorem 8.4.4 for countable G .

Of the three proofs we gave for the finite case of Menger’s theorem, only the last has any chance of being adaptable to the infinite case: the others were by induction on $|G|$ or $|\bigcup \mathcal{P}|$, and both these parameters may now be infinite. The third proof, however, looks more promising: recall that, by Lemmas 3.3.2 and 3.3.3, it provided us with a tool to either find a separator on a given system of A – B paths, or to construct another system of A – B paths that covers more vertices in A and in B .

Lemmas 3.3.2 and 3.3.3 (whose proofs work for infinite graphs too) will indeed form a cornerstone of our proof for Theorem 8.4.4. However, it will not do just to apply these lemmas infinitely often. Indeed, although any finite number of applications of Lemma 3.3.2 leaves us with another system of disjoint A – B paths, an infinite number of iterations may leave nothing at all: each edge may be toggled on and off infinitely often by successive alternating paths, so that no ‘limit system’ of A – B paths will be defined. We shall therefore take another tack: starting at A , we grow simultaneously as many disjoint paths towards B as possible.

To make this precise, we need some terminology. Given a set $X \subseteq V(G)$, let us write $G_{X \rightarrow B}$ for the subgraph of G induced by X and all the components of $G - X$ that meet B .

$G_{X \rightarrow B}$

wave

Let $\mathcal{W} = (W_a \mid a \in A)$ be a family of disjoint paths such that every W_a starts in a . We call $\mathcal{W}$ an $A \rightarrow B$ wave in G if the set Z of final vertices of paths in $\mathcal{W}$ separates A from B in G . (Note that $\mathcal{W}$ may contain infinite paths, which have no final vertex.) Sometimes, we shall wish to consider $A \rightarrow B$ waves in subgraphs of G that contain A but not all of B . For this reason we do not formally require that $B \subseteq V(G)$.

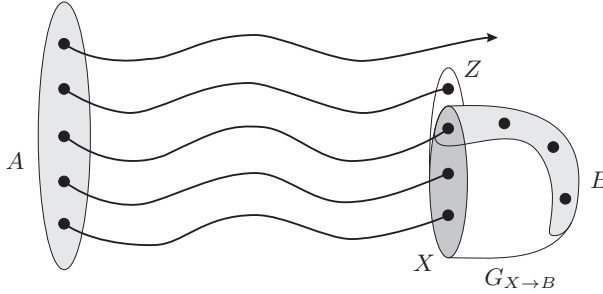


Fig. 8.4.1. A small $A \rightarrow B$ wave $\mathcal{W}$ with boundary X

When $\mathcal{W}$ is a wave, then the set $X \subseteq Z$ of those vertices in Z that either lie in B or have a neighbour in $G_{Z \rightarrow B} - Z$ is a minimal A – B separator in G ; note that $z \in Z$ lies in X if and only if it can be linked to B by a path that has no vertex other than z on $\mathcal{W}$. We call X the *boundary* of $\mathcal{W}$, and often use $(\mathcal{W}, X)$ as shorthand for the wave $\mathcal{W}$ together with its boundary X . If all the paths in $\mathcal{W}$ are finite and $X = Z$, we call the wave $\mathcal{W}$ *large*; otherwise it is *small*. We shall call $\mathcal{W}$ *proper* if at least one of the paths in $\mathcal{W}$ is non-trivial, or if all its paths are trivial but its boundary is a proper subset of A . Every small wave, for example, is proper. Note that while some $A \rightarrow B$ wave always exists, e.g. the family $(\{a\} \mid a \in A)$ of singleton paths, G need not have a proper $A \rightarrow B$ wave. (For example, if A consists of two vertices of $G = K^{10}$ and B of three other vertices, there is no proper $A \rightarrow B$ wave.)

boundary
($\mathcal{W}, X$)large/small
proper $\mathcal{U} + \mathcal{V}$

If $(\mathcal{U}, X)$ is an $A \rightarrow B$ wave in G and $(\mathcal{V}, Y)$ is an $X \rightarrow B$ wave in $G_{X \rightarrow B}$, then the family $\mathcal{W} = \mathcal{U} + \mathcal{V}$ obtained from $\mathcal{U}$ by appending the paths of $\mathcal{V}$ (to those paths of $\mathcal{U}$ that end in X) is clearly an $A \rightarrow B$ wave in G , with boundary Y . Note that $\mathcal{W}$ is large if and only if both $\mathcal{V}$ and $\mathcal{U}$ are large. $\mathcal{W}$ is greater than $\mathcal{U}$ in the following sense.

 $\leq$

Given two path systems $\mathcal{U} = (U_a \mid a \in A)$ and $\mathcal{W} = (W_a \mid a \in A)$, write $\mathcal{U} \leq \mathcal{W}$ if $U_a \subseteq W_a$ for every $a \in A$. Given a chain $(\mathcal{W}^i, X^i)_{i \in I}$ of waves in this ordering, with $\mathcal{W}^i = (W_a^i \mid a \in A)$ say, let $\mathcal{W}^* = (W_a^* \mid a \in A)$ be defined by $W_a^* := \bigcup_{i \in I} W_a^i$. Then $\mathcal{W}^*$ is an $A \rightarrow B$ wave: any A – B path is finite but meets every X^i , so at least one of its vertices lies in X^i for arbitrarily large $(\mathcal{W}^i, X^i)$ and hence is the final vertex of a path in $\mathcal{W}^*$. Clearly $\mathcal{W}^i \leq \mathcal{W}^*$ for all $i \in I$; we call $\mathcal{W}^*$ the *limit* of the waves $\mathcal{W}^i$.

limit wave

As every chain of $A \rightarrow B$ waves is bounded above by its limit wave, Zorn's lemma implies that G has a maximal $A \rightarrow B$ wave $\mathcal{W}$; let X be its boundary. This wave $(\mathcal{W}, X)$ forms the first step in our proof for Theorem 8.4.4: if we can now find disjoint paths in $G_{X \rightarrow B}$ linking all the vertices of X to B , then X will be an A - B separator on these paths preceded by the paths of $\mathcal{W}$ that end in X .

maximal
wave

By the maximality of $\mathcal{W}$, there is no proper $X \rightarrow B$ wave in $G_{X \rightarrow B}$. For our proof it will thus suffice to prove the following (renaming X as A):

Lemma 8.4.5. *If G has no proper $A \rightarrow B$ wave, then G contains a set of disjoint A - B paths linking all of A to B .*

Our approach to the proof of Lemma 8.4.5 is to enumerate the vertices in $A =: \{a_1, a_2, \dots\}$, and to find the required A - B paths $P_n = a_n \dots b_n$ in turn for $n = 1, 2, \dots$. Since our premise in Lemma 8.4.5 is that G has no proper $A \rightarrow B$ wave, we would like to choose P_1 so that $G - P_1$ has no proper $(A \setminus \{a_1\}) \rightarrow B$ wave: this would restore the same premise to $G - P_1$, and we could proceed to find P_2 in $G - P_1$ in the same way.

$a_1, a_2, \dots$
 P_n

We shall not be able to choose P_1 quite like this, but we shall be able to do something almost as good. We shall construct P_1 so that deleting it (as well as a few more vertices outside A) leaves a graph that has a large maximal $(A \setminus \{a_1\}) \rightarrow B$ wave $(\mathcal{W}, A')$. We then earmark the paths $W_n = a_n \dots a'_n$ ($n \geq 2$) of this wave as initial segments for the paths P_n . By the maximality of $\mathcal{W}$, there is no proper $A' \rightarrow B$ wave in $G_{A' \rightarrow B}$. In other words, we have restored our original premise to $G_{A' \rightarrow B}$, and can find there an A' - B path $P'_2 = a'_2 \dots b_2$. Then $P_2 := a_2 W_2 a'_2 P'_2$ is our second path for Lemma 8.4.5, and we continue inductively inside $G_{A' \rightarrow B}$.

Given a set $\hat{A}$ of vertices in G , let us call a vertex $a \notin \hat{A}$ *linkable* for $(G, \hat{A}, B)$ if $G - \hat{A}$ contains an a - B path P and a set $X \supseteq V(P)$ of vertices such that $G - X$ has a large maximal $\hat{A} \rightarrow B$ wave. (The first such a we shall be considering will be a_1 , and $\hat{A}$ will be the set $\{a_2, a_3, \dots\}$.)

linkable

Lemma 8.4.6. *Let $a^* \in A$ and $\hat{A} := A \setminus \{a^*\}$, and assume that G has no proper $A \rightarrow B$ wave. Then a^* is linkable for $(G, \hat{A}, B)$.*

Proof of Lemma 8.4.5 (assuming Lemma 8.4.6). Let G be as in Lemma 8.4.5, i.e. assume that G has no proper $A \rightarrow B$ wave. We construct subgraphs $G_1, G_2, \dots$ of G satisfying the following statement (Fig. 8.4.2):

G_n contains a set $A^n = \{a_n^n, a_{n+1}^n, a_{n+2}^n, \dots\}$ of distinct vertices such that G_n has no proper $A^n \rightarrow B$ wave. In G there are disjoint paths P_i ($i < n$) and W_i^n ($i \geq n$) starting at a_i . The P_i are disjoint from G_n and end in B . The W_i^n end in a_i^n and are otherwise disjoint from G_n . (*)

Clearly, the paths $P_1, P_2, \dots$ will satisfy Lemma 8.4.5.

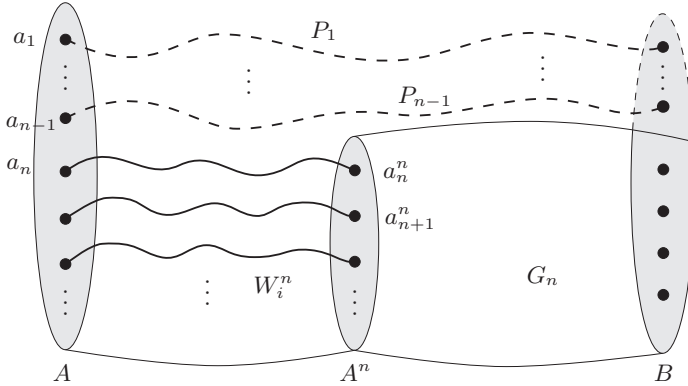


Fig. 8.4.2. G_n has no proper $A^n \rightarrow B$ wave

Let $G_1 := G$, and put $a_i^1 := a_i$ and $W_i^1 := \{a_i\}$ for all $i \geq 1$. Since by assumption G has no proper $A \rightarrow B$ wave, these definitions satisfy (*) for $n = 1$. Suppose now that (*) has been satisfied for n . Put $\hat{A}^n := A^n \setminus \{a_n^n\}$. By Lemma 8.4.6 applied to G_n , we can find in $G_n - \hat{A}^n$ an $a_n^n - B$ path P and a set $X_n \supseteq V(P)$ such that $G_n - X_n$ has a large maximal $\hat{A}^n \rightarrow B$ wave $(\mathcal{W}, A^{n+1})$. Let P_n be the path $W_n^n \cup P$. For $i \geq n + 1$, let W_i^{n+1} be W_i^n followed by the path of $\mathcal{W}$ starting at a_i^n , and call its last vertex a_i^{n+1} . By the maximality of $\mathcal{W}$ there is no proper $A^{n+1} \rightarrow B$ wave in $G_{n+1} := (G_n - X_n)_{A^{n+1} \rightarrow B}$, so (*) is satisfied for $n + 1$. $\square$

To complete our proof of Theorem 8.4.4, it remains to prove Lemma 8.4.6. For this, we need another lemma:

Lemma 8.4.7. *Let x be a vertex in $G - A$. If G has no proper $A \rightarrow B$ wave but $G - x$ does, then every $A \rightarrow B$ wave in $G - x$ is large.*

(3.3.2) *Proof.* Suppose $G - x$ has a small $A \rightarrow B$ wave $(\mathcal{W}, X)$. Put $B' := X \cup \{x\}$, and let $\mathcal{P}$ denote the set of $A - X$ paths in $\mathcal{W}$ (Fig. 8.4.3). If G contains an $A - B'$ separator S on $\mathcal{P}$, then replacing in $\mathcal{W}$ every $P \in \mathcal{P}$ with its initial segment ending in S we obtain a small (and hence proper) $A \rightarrow B$ wave in G , which by assumption does not exist. By Lemmas 3.3.3

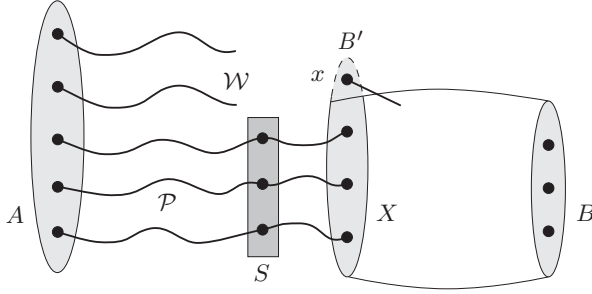


Fig. 8.4.3. A hypothetical small $A \rightarrow B$ wave in $G - x$

and 3.3.2, therefore, G contains a set $\mathcal{P}'$ of disjoint $A-B'$ paths exceeding $\mathcal{P}$. The set of last vertices of these paths contains X properly, and hence must be all of $B' = X \cup \{x\}$. But B' separates A from B in G , so we can turn $\mathcal{P}'$ into an $A \rightarrow B$ wave in G by adding as singleton paths any vertices of A it does not cover. As x lies on $\mathcal{P}'$ but not in A , this is a proper wave, which by assumption does not exist. $\square$

Proof of Lemma 8.4.6. We inductively construct trees $T_0 \subseteq T_1 \subseteq \dots$ in $G - (\hat{A} \cup B)$ and path systems $\mathcal{W}_0 \leq \mathcal{W}_1 \leq \dots$ in G so that each $\mathcal{W}_n$ is a large maximal $\hat{A} \rightarrow B$ wave in $G - T_n$.

Let $\mathcal{W}_0 := (\{a\} \mid a \in \hat{A})$. Clearly, $\mathcal{W}_0$ is an $\hat{A} \rightarrow B$ wave in $G - a^*$, and it is large and maximal: if not, then $G - a^*$ has a proper $\hat{A} \rightarrow B$ wave, and adding the trivial path $\{a^*\}$ to this wave turns it into a proper $A \rightarrow B$ wave (which by assumption does not exist). If $a^* \in B$, the existence of $\mathcal{W}_0$ makes a^* linkable for $(G, \hat{A}, B)$. So we assume that $a^* \notin B$. Now $T_0 := \{a^*\}$ and $\mathcal{W}_0$ are as desired.

Suppose now that T_n and $\mathcal{W}_n$ have been defined, and let A_n denote the set of last vertices of the paths in $\mathcal{W}_n$. Since $\mathcal{W}_n$ is large, A_n is its boundary, and since $\mathcal{W}_n$ is maximal, $G_n := (G - T_n)_{A_n \rightarrow B}$ has no proper $A_n \rightarrow B$ wave (Fig. 8.4.4).

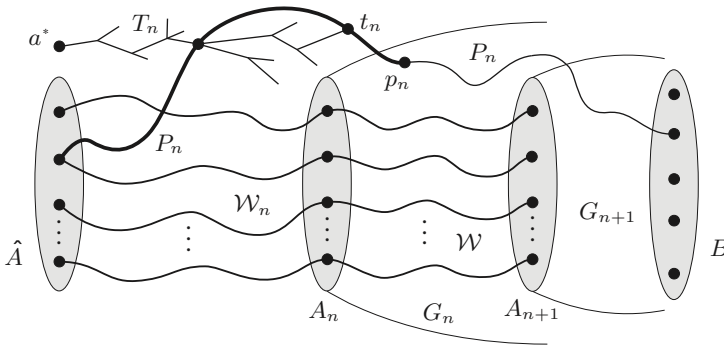


Fig. 8.4.4. As $\mathcal{W}_n$ is maximal, G_n has no proper $A_n \rightarrow B$ wave

Note that A_n does not separate A from B in G : if it did, then $\mathcal{W}_n \cup \{a^*\}$ would be a small $A \rightarrow B$ wave in G , which does not exist. Hence, $G - A_n$ contains an A - B path P , which meets T_n because $(\mathcal{W}_n, A_n)$ is a wave in $G - T_n$. Let $P =: P_n$ be chosen so that its vertex p_n following its last vertex t_n in T_n is minimal in some fixed enumeration of $V(G)$.

Let $T_{n+1} := T_n \cup t_n P_n p_n$ be the tree obtained from T_n by adding the edge $t_n p_n$. If $G_n - p_n$ has no proper $A_n \rightarrow B$ wave, then $\mathcal{W}_{n+1} := \mathcal{W}_n$ is large and maximal not only in $G - T_n$ but also in $G - T_{n+1}$. If $G_n - p_n$ has a proper $A_n \rightarrow B$ wave, $\mathcal{W}$ say, we choose it maximal. Since G_n has no such wave, $\mathcal{W}$ is large by Lemma 8.4.7. Now $\mathcal{W}_{n+1} := \mathcal{W}_n + \mathcal{W}$ is a large maximal $A_n \rightarrow B$ wave in $G - T_{n+1}$. In either case T_{n+1} and $\mathcal{W}_{n+1}$ are as desired, unless $p_n \in B$. But then a^* is linkable for $(G, \hat{A}, B)$ with a^* - B path $a^* T_{n+1} p_n$, wave $\mathcal{W}_{n+1}$ and $X = V(T_{n+1})$.

Put $T^* := \bigcup_{n \in \mathbb{N}} T_n$. Then the $\mathcal{W}_n$ are $\hat{A} \rightarrow B$ waves also in $G - T^*$; let $(\mathcal{W}^*, A^*)$ be their limit. Our aim is to show that A^* separates A from B not only in $G - T^*$ but even in G : then $(\mathcal{W}^* \cup \{a^*\}, A^*)$ is a small $A \rightarrow B$ wave in G , which by assumption does not exist: a contradiction.

Suppose there exists an A - B path Q in $G - A^*$. Let t be its last vertex in T^* . Since T^* does not meet B , there is a vertex p following t on Q . This p is not among the p_n , since T^* contains those but not p . Let n be large enough that $t \in T_n$ and that p precedes p_n in our fixed enumeration of $V(G)$. (As $p_n \in T_{n+1} - T_n$, the p_n are pairwise distinct.)

The fact that $a^* T_n t Q$ was not chosen as P_n means that its portion pQ outside T_n meets A_n , say in a vertex q . Now $q \notin A^*$ by the choice of Q . Let W be the path in $\mathcal{W}_n$ that joins $\hat{A}$ to q ; this path too avoids A^* . But then WqQ contains an $\hat{A}$ - B path in $G - T^*$ avoiding A^* , which contradicts the definition of A^* . $\square$

The proof of Theorem 8.4.4 for countable G is now complete.

Turning now to matching, let us begin with a simple problem that is intrinsically infinite. Given two sets A, B and injective functions $A \rightarrow B$ and $B \rightarrow A$, is there necessarily also a bijection between A and B ? Indeed there is – this is the famous Cantor-Bernstein theorem from elementary set theory. Recast in terms of matchings, the proof becomes very simple:

Proposition 8.4.8. *Let G be a bipartite graph, with bipartition $\{A, B\}$ say. If G contains a matching of A and a matching of B , then G has a 1-factor.*

Proof. Let H be the multigraph on $V(G)$ whose edge set is the disjoint union of the two matchings. (Thus, any edge that lies in both matchings becomes a double edge in H .) Every vertex in H has degree 1 or 2. In fact, it is easy to check that every component of H is an even cycle or an infinite path. Picking every other edge from each component, we obtain a 1-factor of G . $\square$

The corresponding path problem in non-bipartite graphs, with sets of disjoint A – B paths instead of matchings, is less trivial. Let us say that a set $\mathcal{P}$ of paths in G *covers* a set U of vertices if every vertex in U is an endvertex of a path in $\mathcal{P}$. *covers*

Theorem 8.4.9. (Pym 1969)

Let G be a graph, and let $A, B \subseteq V(G)$. Suppose that G contains two sets of disjoint A – B paths, one covering A and one covering B . Then G contains a set of disjoint A – B paths covering $A \cup B$.

Some hints for a proof of Theorem 8.4.9 are included with Exercise 70.

Next, let us see how the standard matching theorems for finite graphs – König, Hall, Tutte, Gallai-Edmonds – extend to infinite graphs. For locally finite graphs, they all have straightforward extensions by compactness; see Exercises 26–29. But there are also very satisfactory extensions to graphs of arbitrary cardinality. Their proofs form a coherent body of theory and are much deeper, so we shall only be able to state those results and point out how some of them are related. But, as with Menger’s theorem, the statements themselves are interesting too: finding the ‘right’ restatement of a given finite result to make a substantial infinite theorem is by no means easy, and most of them were found only as the theory itself developed over the years.

Let us start with bipartite graphs. The following Erdős-Menger-type extension of König’s theorem (2.1.1) is now a corollary of Theorem 8.4.4:

Theorem 8.4.10. (Aharoni 1984)

Every bipartite graph has a matching, M say, and a vertex cover of its edge set that consists of exactly one vertex from every edge in M .

What about an infinite version of the marriage theorem (2.1.2)? The finite theorem says that a matching exists as soon as every subset S of the first partition class has enough neighbours in the second. But how do we measure ‘enough’ in an infinite graph? Just as in Menger’s theorem, comparing cardinalities is not enough (Exercise 27).

However, there is a neat way of rephrasing the marriage condition for a finite graph without appealing to cardinalities. Call a subset X of one partition class *matchable* to a subset Y of the other if the subgraph spanned by X and Y contains a matching of X . Now if S is *minimal* with $|S| > |N(S)|$ then, by the marriage theorem, S is ‘larger’ than $N(S)$ also in the sense that S is not matchable to $N(S)$ but $N(S)$ is matchable to S . (Indeed, by the minimality of S and the marriage theorem, any $S' \subseteq S$ with $|S'| = |S| - 1$ can be matched to $N(S)$. As $|S'| = |S| - 1 \geq |N(S)|$, this matching covers $N(S)$.) Thus, if there is any obstruction S of the type $|S| > |N(S)|$ to a perfect matching, there is also one where S is larger than $N(S)$ in this other sense: that S is not matchable to $N(S)$ but $N(S)$ is matchable to S . *matchable*